

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

.Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch: 1
Date of Assignment: 4/08/2026	Date/Time of Submission: 4/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code:-

Manifest.json:-

```
{
  "name": "College Information Portal",
  "short_name": "College Portal",
  "description": "College website for notices, timetables and study materials.",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon3.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon4.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

service-worker.js:-

```
const cache = "college-pwa-v1";
self.addEventListener("install", event => {
  event.waitUntil(
    caches.open(cache).then(cacheData => {
      return cacheData.addAll([
        "/",
        "/offline.html"
      ]);
    })
  );
})
```

```
);  
  console.log("Service Worker Installed");  
});  
self.addEventListener("fetch", e => {  
  e.respondWith(  
    fetch(e.request).catch(() => {  
      return caches.match("/offline.html");  
    })  
  );  
});
```

Offline.html:-

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Offline College Portal</title>  
</head>  
<body>  
  <h1>College Information Portal</h1>  
  <h2>Offline Mode</h2>  
  <p>You are currently offline.</p>  
  <h3>Available Information</h3>  
  <ul>  
    <li>About College</li>  
    <li>Courses Offered</li>  
    <li>Departments</li>  
    <li>Faculty Details</li>  
    <li>Contact Information</li>  
  </ul>  
  <p>Please reconnect to the internet for the latest updates.</p>  
</body>  
</html>
```

Output:-

College Information Portal

Offline Mode

You are currently offline.

Available Information

- About College
- Courses Offered
- Departments
- Faculty Details
- Contact Information

Please reconnect to the internet for the latest updates.

b. Create a Student Record Management PWA.

Code:-

[service-worker.js](#):-

```
const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});
self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
      return response || fetch(e.request);
    })
  );
});
```

manifest.json:-

```
{
  "name": "college portal",
```

```
"short_name": "cp",
"start_url": "/",
"display": "standalone",
"icons": [
  {
    "src": "icons/icon1.png",
    "sizes": "148x148",
    "type": "image/png",
    "purpose": "any"
  },
  {
    "src": "icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png",
    "purpose": "any"
  }
]
```

HomeController.cs:-

```
public IActionResult Index()
{
    student s = new student();
    s.RollNo = 5;
    s.Name = "Ritesh";
    return View(s);
}
```

Index.cshtml:-

```
<h1>Offline Page</h1>
<h2>Student Record</h2>
<p>Roll No : @Model.RollNo</p>
<p>Name : @Model.Name</p>
```

Student.cs:-

```
namespace WebApplication12.Models
{
    public class student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

Output:-

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page

Student Record

Roll No : 5

Name : Ritesh

[WebApplication12](#) 

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Welcome

Learn about [building Web apps with ASP.NET Core](#).

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page

Student Record

Roll No : 5

Name : Ritesh

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